

State and proof roadmap for **Lemmas**

Quantitative strict-AT formalization

Snapshot checked on 1 August 2026

1 Executive status

The target **LatalaMeetsAT** builds successfully with Lean 4.32.1. The latest check completed 3152 jobs. The top-level source now requests a uniform **FreeEnergyEndpointBridge** instance and uses it to recover the public **rs_freeEnergy_error** theorem from the proved conditional result **rs_freeEnergy_error_of_endpoint_bridge**.

The directory **Lemmas** now contains 24 Lean files. Nineteen files have no local **sorry**; five files contain six occurrences in total. There are no project-specific **axiom** declarations and no **admits** in this directory. The six placeholders comprise one definition, **cavityInterpolatedAverage**, and five theorems.

The placeholder count understates the unfinished analytic work. Several former placeholders have been replaced by explicit propositions or typeclasses: **HasSmartPathCovarianceDerivative**, **SpecializedGTInterpolation**, **GTFunctionalCoercivity**, **HasFreeEnergyEndpointBridge**, **FreeEnergyEndpointBridge**, and **HasCavityRemainderBound**. The theorems using these interfaces are locally **sorry**-free, but the repository does not construct the required instances or hypotheses. They are therefore conditional results, not unconditional certifications of the paper.

The successful build reports that **SpinGlass.AT.quantitative_strictAT** depends on **sorryAx**, along with the ordinary foundations **propext**, **Classical.choice**, and **Quot.sound**. Thus the source is structurally build-complete but is not certified.

Since the preceding snapshot, the singular Latala reference-mean identity has been proved, **scalarOrderParameter** has received explicit Gaussian formulas, and the elementary GT operators plus the explicit finite recursion have been consolidated into **GT/Interpolation.lean**. The final free-energy estimate has also been proved from an explicit endpoint-bridge hypothesis. The former files **GT/Semigroup.lean** and **GT/HalfCascade.lean** have been removed.

2 Import spine

The principal dependency spine is now

```
Replicas → RSPParameters → UniformData → SmartPath → GaussianDerivative →  
FreeEnergyDerivative → Scalar/Semigroup → Scalar/LatalaKernel → Scalar/StrictATSign  
→ GT/Defs → GT/Interpolation → GT/Coercivity → CoupledPressure → FixedDeviation →  
Absorption → FreeEnergy → Replicon → CompactCorollary → MainResult.
```

The cavity branch starts independently at **SmartPath**. Definitions feed both interpolation and coefficients; coefficients feed stability and the system; absorption joins stability, the system, and fixed-deviation estimates.

3 State of every Lean file

File	Local state	What is present or missing
<code>Replicas.lean</code>	locally complete	Finite configurations, positive partition functions, Gibbs-weight bounds and normalization, permutation invariance, overlap bounds, linearity, monotonicity, and the mixed-overlap estimate.
<code>RSPParameters.lean</code>	locally complete	Gaussian RS quantities, the selected fixed point, positivity and range of q , identities for A , and anomalous-versus-replicon comparison.
<code>UniformData.lean</code>	locally complete	Compact-uniform parameter package and the strict-AT path gap.
<code>SmartPath.lean</code>	locally complete	Covariance kernel, abstract centered Gaussian path, full Hamiltonian with external field, overlap second moment, and path gap. It still has no explicit finite Gaussian coordinate realization.
<code>GaussianDerivative.lean</code>	conditional	The covariance operator is explicit and the public theorem is <code>sorry-free</code> , but it merely projects the derivative field of an unconstructed <code>HasSmartPathCovarianceDerivative</code> instance.
<code>FreeEnergyDerivative.lean</code>	open (1)	Definitions and <code>rsGap_deriv</code> are complete; <code>smartPath_freeEnergy_deriv</code> remains a placeholder and does not yet request the new covariance-derivative instance explicitly.
<code>Scalar/Semigroup.lean</code>	locally complete	Both branches of <code>scalarPsi</code> and <code>scalarTrialValue_eq</code> are proved, including Gaussian integrability and convolution.
<code>Scalar/LatalaKernel.lean</code>	locally complete	Derivative and sign of <code>latalaH</code> , antitonicity, the opposite-monotonicity covariance inequality, singular reference normalization, and the weighted kernel comparison are proved.
<code>Scalar/StrictATSign.lean</code>	open (1)	<code>scalarOrderParameter</code> is now an explicit piecewise Gaussian formula. Differentiation, diffusion comparison, strict sign, and compact-uniform local constants remain.
<code>GT/Defs.lean</code>	locally complete	Attainable overlaps, constrained free energy, signed matrix path, mass profile, correction, terminal condition, and endpoint identities.
<code>GT/Interpolation.lean</code>	conditional	Contains the zero, half, and one operators, the constant half-mass identity, explicit Gaussian recursion steps, <code>gtSemigroupSolution</code> , and <code>gtFunctional</code> . The two-replica bound is only a projection from the unconstructed <code>SpecializedGTInterpolation</code> class.
<code>GT/Coercivity.lean</code>	conditional	The local constant, abstract Taylor-loss lemma, overlap-range lemma, and composition theorem are proved. The analytic functional estimate is packaged as the unconstructed <code>GTFunctionalCoercivity</code> class.
<code>CoupledPressure.lean</code>	open (1)	Correct quadratic pressure definitions are present. The uniform sublinear overlap estimate remains; its public alias is complete.
<code>FixedDeviation.lean</code>	open (1)	Exponentially small fixed-deviation tails remain.

File	Local state	What is present or missing
<code>Cavity/Defs.lean</code>	locally complete	The observables A, B, C , the identity between A and the overlap second moment, cavity vector, source vector, and 3×3 cavity matrix.
<code>Cavity/Interpolation.lean</code>	open (2)	The elementary third-moment bound and compatibility name are complete. The last-spin interpolation definition and its second-derivative estimate remain.
<code>Cavity/Coefficients.lean</code>	locally complete	Replica-edge classification, decoupled coefficients, and equality with the displayed cavity matrix.
<code>Cavity/System.lean</code>	conditional	No local placeholder. The system is algebraic because the remainder is defined as the residual. Its analytic bound is a predicate and is assumed, not constructed.
<code>Cavity/Stability.lean</code>	locally complete	Explicit matrix inverse with two-sided identities, triangular determinant formula, anomalous and replicon gaps, uniform inverse estimate, and replicon left eigenvector.
<code>Absorption.lean</code>	locally complete	Pointwise cubic splitting, exact cavity-vector norm, domination of B and C by A , and the uniform NA_s bound are proved from the stated cavity-remainder hypothesis.
<code>FreeEnergy.lean</code>	conditional	The free-energy estimate is proved using the mean-value theorem, positivity, and absorption. A public wrapper consumes the unconstructed uniform class <code>FreeEnergyEndpointBridge</code> .
<code>Replicon.lean</code>	locally complete	Third-moment little- o and replicon susceptibility are proved from earlier tail, absorption, stability, and assumed cavity-remainder results.
<code>CompactCorollary.lean</code>	locally complete	Compactness plus continuity and strict AT produce <code>UniformATData</code> .
<code>MainResult.lean</code>	conditional	The assembly builds and combines absorption, free energy, and replicon results. Its context assumes <code>FreeEnergyEndpointBridge</code> ; its argument assumes the existence of a cavity-remainder bound. Transitive placeholders still yield <code>sorryAx</code> .

4 Remaining work and recommended prover

The rating concerns the current source interfaces. “Codex” should lead definition design, API repair, supporting lemmas, and cross-file integration. “Aristotle” is useful for a focused Lean proof only after the relevant interfaces and mathematical estimates have stable statements.

File or contract	Difficulty	Better lead	Reason and possible handoff
Gaussian derivative contract	difficult	Codex	The operator is written down, but an instance of <code>HasSmartPathCovarianceDerivative</code> still requires finite-dimensional Gaussian covariance interpolation or an explicit affine realization with domination. Hand off the final finite-sum algebra after the analytic API is fixed.

File or contract	Difficulty	Better lead	Reason and possible handoff
<code>FreeEnergyDerivative.lean</code>	difficult	Codex	The proof must consume the Gaussian instance, derive the diagonal correction and covariance cancellation, and normalize replica sums. Focused algebraic sublemmas may then be handed off.
<code>Scalar/StrictATSign.lean</code>	difficult	Codex	The formula is now explicit, so the remaining work is differentiation under Gaussian integrals, the two branch estimates, and uniform constants.
GT interpolation contract	difficult	Codex	The explicit recursion exists, but constructing <code>SpecializedGTInterpolation</code> still requires the finite-cascade interpolation, endpoint identifications, and derivative sign.
GT coercivity contract	difficult	Codex	Constructing <code>GTFunctionalCoercivity</code> requires multiplier derivatives, a uniform second-derivative estimate, the strict scalar sign, and signed far-from- q bounds.
<code>CoupledPressure.lean</code>	difficult	Codex	The argument combines GT coercivity, overlap discretization, Gaussian concentration, convexity, and Gronwall, and its statement will need the relevant contracts.
<code>FixedDeviation.lean</code>	difficult	Codex	The covariance-only disorder still lacks the explicit coordinate or direct concentration interface needed for the gradient estimate.
<code>Cavity/Interpolation.lean</code>	difficult	Codex	The last-spin decomposition is absent, and the derivative expansion requires replica book-keeping, endpoint identities, and uniform estimates.
Free-energy bridge	moderate	Codex	The final calculus argument is complete. What remains for <code>HasFreeEnergyEndpointBridge</code> is continuity of the covariance-dependent free energy and the $s = 0$ product endpoint identity.
Top-level certification	moderate	Codex	The assembly now builds. A final theorem must construct the endpoint bridge and cavity-remainder bound, and later propagate any Gaussian or GT contracts exposed when downstream placeholders are removed.

5 The six remaining sorry obligations

5.1 Free-energy derivative

1. In `FreeEnergyDerivative.lean`, prove `smartPath_freeEnergy_deriv`. Instantiate the finite-Gaussian covariance derivative, prove the diagonal correction and cancellation, and rewrite the replica sums as `overlapSecondMoment`. The target derivative is $\frac{\beta^2}{4}((1 - q)^2 - \mathbb{E}\langle (R_{12} - q)^2 \rangle)$.

5.2 Scalar strict sign

2. `Scalar/StrictATSign.lean:strictAT_sign`. Differentiate the two explicit formulas for `scalarOrderParameter`. Below q , use conditional Jensen and the path gap. Above q , use the proved Latała comparison. Then obtain the uniform local constants by continuity and compactness.

5.3 Coupled pressure and concentration

3. `CoupledPressure.lean:coupledPressure_sublinear`. Split over the $N + 1$ attainable overlaps, apply the GT functional contracts and Gaussian concentration, then combine the pressure envelope with convexity, `rsGap_deriv`, and Gronwall to obtain a uniform sequence tending to zero.
4. `FixedDeviation.lean:fixedDeviation`. Prove the uniform $O_K(N)$ squared-gradient estimate for the quadratic exponential moment and apply a finite-Gaussian concentration theorem compatible with `RSSmartPathDisorder`.

5.4 Cavity interpolation

5. `Cavity/Interpolation.lean:cavityInterpolatedAverage`. Define the last-spin interpolation $\nu_{s,u}(F)$ from an explicit last-spin Gaussian decomposition, with normalization, measurability, and both endpoint identities.
6. `Cavity/Interpolation.lean:cavity_secondDerivative_bound`. Differentiate the interpolation twice, classify replica-edge terms through `Cavity/Coefficients`, and bound the remainder by $C_K(\text{thirdMoment} + N^{-3})$.

6 Unconstructed contracts without local sorry

Gaussian covariance derivative. `HasSmartPathCovarianceDerivativepath` contains exactly the derivative theorem previously represented by a placeholder. No instance is supplied.

GT interpolation. `SpecializedGTInterpolation` contains the finite-volume two-replica GT bound for every multiplier and attainable overlap. No instance is supplied.

GT functional coercivity. `GTFunctionalCoercivitydata` contains the analytic quadratic loss of the explicit functional. No instance is supplied.

Free-energy endpoints. `HasFreeEnergyEndpointBridgepath` packages continuity on $[0, 1]$ and the $s = 0$ product identity. The uniform class `FreeEnergyEndpointBridge` supplies this proposition for every positive-size path. The top-level theorem assumes the class, but the repository supplies no instance.

Cavity remainder. `HasCavityRemainderBounddataC` remains the analytic input to absorption and the final conclusions. The algebraic cavity system does not prove it.

These contracts are useful dependency markers, but moving a claim from `sorry` into an assumed proposition does not complete its proof.

7 Recommended proof roadmap

Model and Gaussian API. Construct `HasSmartPathCovarianceDerivative` from a finite-index Gaussian-law module or strengthen the smart path with an explicit affine realization. Use the same model for concentration and last-spin decomposition. Then finish the free-energy derivative.

Scalar package. Prove `strictAT_sign` from the explicit lower and upper Gaussian formulas. The convolution identity and all Latala kernel lemmas needed by this stage are now complete.

GT package. Construct `SpecializedGTInterpolation` for the explicit recursion, then construct `GTFunctionalCoercivity`. Expose multiplier derivatives and factorization as reusable named lemmas.

Pressure and tails. Finish the sublinear pressure and fixed-deviation theorems using the common finite-Gaussian concentration interface.

Cavity package. Define the last-spin interpolation, prove its second-derivative bound, and use it to construct `HasCavityRemainderBound`. This is the missing analytic connection between the cavity expansion and absorption.

Free-energy endpoints. Construct `FreeEnergyEndpointBridge` using covariance-law continuity and the independent one-site endpoint. The $O(N^{-1})$ estimate and its top-level assembly are already complete from that class.

Certification. Run `lakebuildLatalaMeetsAT`, search the Lean sources for `sorry`, `admit`, and project-specific `axiom`, and inspect `\#printaxiomsSpinGlass.AT.quantitative_strictAT`. Certification requires a successful build and absence of `sorryAx`; it also requires that the public theorem no longer obtains its analytic content from unproved assumptions.

8 Structural issues

Cavity system. The theorem `cavity_system` follows by unfolding the definition of `cavityRemainder`. Its analytic content is the separate, unconstructed remainder bound.

- In `GaussianDerivative.lean`, the theorem `quenchedGibbs_deriv_of_covariance_deriv` is a wrapper around a typeclass field. It does not yet prove Gaussian interpolation from the covariance assumptions in `SmartPath.lean`.
- `GT/Interpolation.lean:twoReplica_GT_bound` and `GT/Coercivity.lean:gt_quadratic_coercivity` are conditional composition theorems. The explicit recursion has removed the former definitional placeholder, but neither core analytic estimate has been proved.
- `FreeEnergy.lean` no longer contains a placeholder, but its public theorem obtains the endpoint facts from the assumed `FreeEnergyEndpointBridge` class.
- `MainResult.lean:quantitative_strictAT` assumes `FreeEnergyEndpointBridge` in its context and exposes an existential cavity-remainder hypothesis. The statement remains conditional until both inputs are constructed from the smart-path model and compact strict-AT data.
- `SmartPath.lean` specifies Gaussian marginals and covariance but no concrete coordinate realization. This single model-level gap blocks the Gaussian derivative instance, direct concentration, endpoint-law continuity, and the last-spin decomposition.